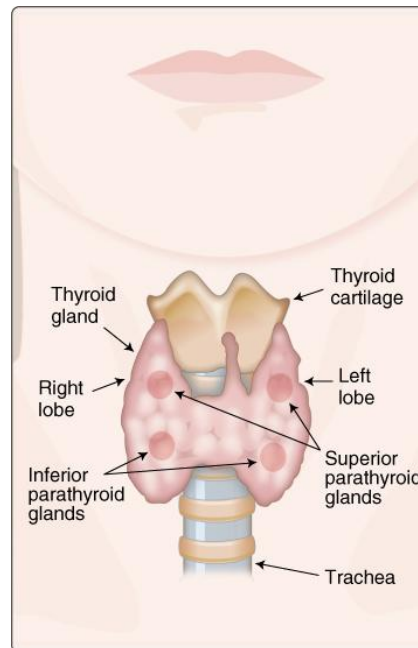


Basic Principles of Medicine 2

Module : Endocrinology & Reproduction

ER Lecture 09

Calcium homeostasis



Presented by
Dr. Alvin Chitterman MD, MPH
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Recommended reading

- Rhoades RA, Bell DR. Medical Physiology: Principles for Clinical Medicine, 6e, Chapter 35, pages 759-772. [Endocrine Regulations of Calcium, Phosphate, and Bone Homeostasis | Medical Physiology: Principles for Clinical Medicine, 6e | Medical Education | Health Library \(oclc.org\)](#)
- Costanzo, L. S. (2021). *Costanzo Physiology* (7th ed.). Elsevier Limited (UK)., 449-460 [ClinicalKey Student: Costanzo Physiology \(elsevier.com\)](#)
- Slides with blue background are the text version of the explanations given in the classroom and thus, might not be showed during the lecture. However, the content MUST be read and understood.

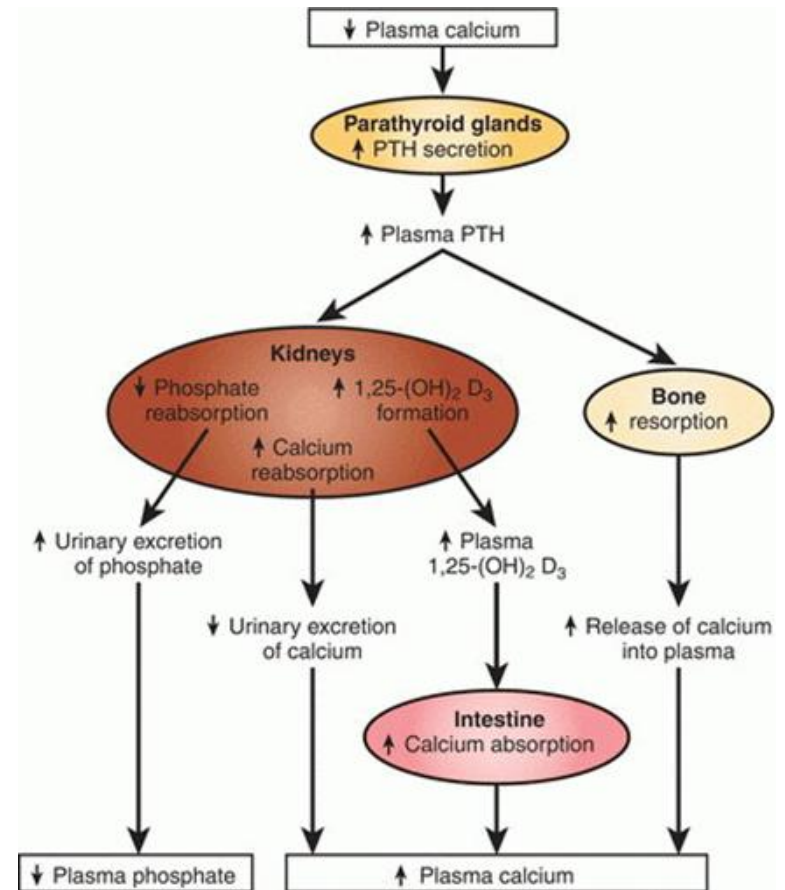
Objectives

SOM.MK.II.BPM2.1.ER.1. PHYS.1056	Identify the normal range of dietary calcium intake, calcium distribution in the body, and routes of calcium excretion.
SOM.MK.II.BPM2.1.ER.1. PHYS.1057	Identify the normal range of dietary phosphate intake, phosphate distribution in the body, and routes of phosphate excretion.
SOM.MK.II.BPM2.1.ER.1. PHYS.1058	Describe the organ that secretes parathyroid hormone, its chemical nature and its mechanism of action
SOM.MK.II.BPM2.1.ER.1. PHYS.1059	Describe the regulation of parathyroid hormone secretion and the role of the calcium sensing receptor.
SOM.MK.II.BPM2.1.ER.1. PHYS.1060	Identify the sources of vitamin D and diagram the biosynthetic pathway and the organs involved in modifying it to the biologically active 1,25(OH) ₂ D ₃ (1-25 dihydroxy cholecalciferol).
SOM.MK.II.BPM2.1.ER.1. PHYS.1061	Identify the target organs and cellular mechanisms of action for vitamin D.
SOM.MK.II.BPM2.1.ER.1. PHYS.1062	Describe calcitonin and its role in humans
SOM.MK.II.BPM2.1.ER.1. PHYS.1063	Explain the clinically relevant consequences of oversecretion undersecretion of parathyroid hormone.
SOM.MK.II.BPM2.1.ER.1. PHYS.1064	Describe the consequences of vitamin D deficiency and vitamin D excess.

The Big Picture

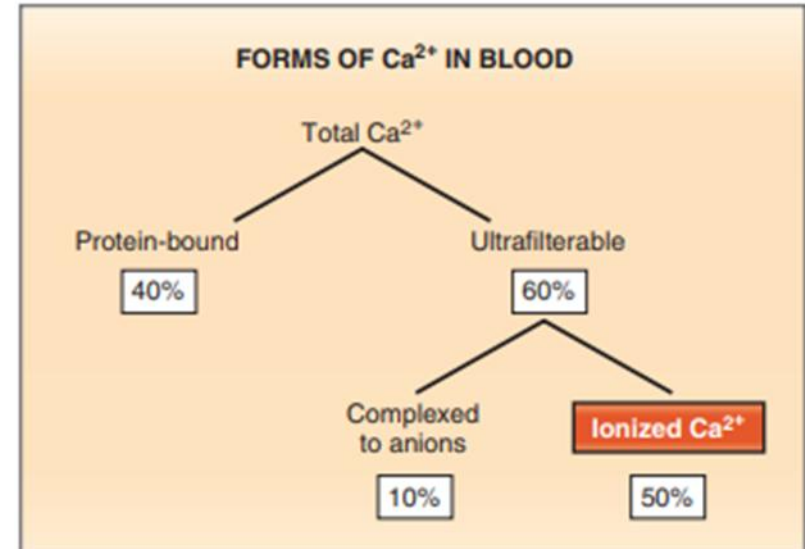
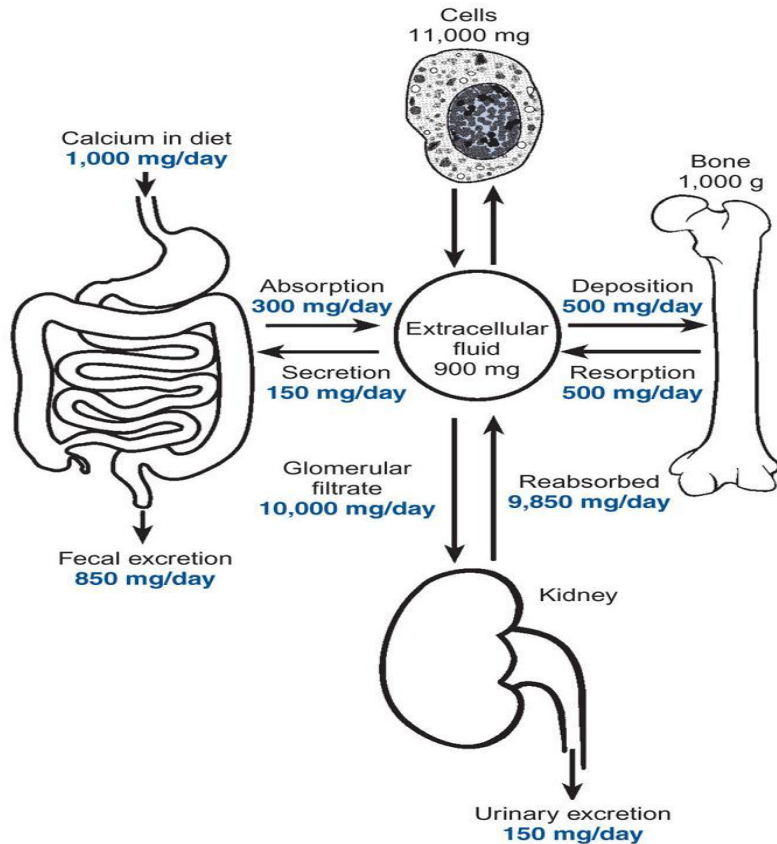
Calcium distribution & balance

- Ca^{2+} plays a central role in regulation of important cellular functions. The plasma concentration is **8.6 -10 mg/dl**
- It accounts for 2% of the body weight. 99% Ca^{2+} is in bone & 1 % is in body fluids.



Note: The ionized form is physiologically active. Both ICF & ECF Ca^{2+} in the plasma is tightly regulated

Typical daily exchanges of calcium



When Ca^{2+} deficiency occurs
90% is absorbed from GIT

Calcium source & balance

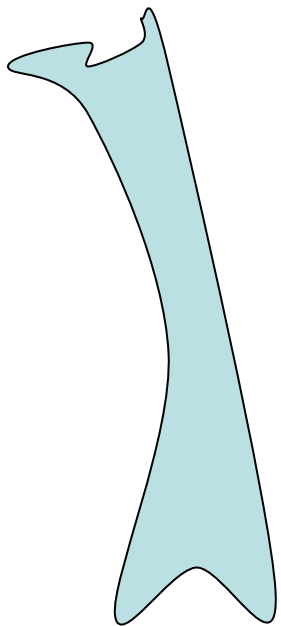
- Major source of Ca^{2+} in the body is diet & bone. Milk, cheese & eggs are rich source of Ca^{2+}
- If diet is deficient in Ca^{2+} it is absorbed from bone
- Kidney plays a major role in regulation of Ca^{2+} along with the bone
- Gut absorption, bone uptake & release, kidney Ca^{2+} reabsorption are tightly regulated to maintain Ca^{2+} within physiologic range

Nursing & Pregnant mothers

- Nursing & pregnant mothers have high Ca^{2+} requirements
- About 625 mmols/day is absorbed via placenta for fetus, in pregnant mothers. Infants receive 2000 mmols/day via milk
- Fetus and infant, both need it for bone formation
- Ca^{2+} deficiency can occur in pregnant & nursing mothers

Bone composition

Ca²⁺ in bone exists in 2 pools



Readily exchangeable pool (small reservoir)

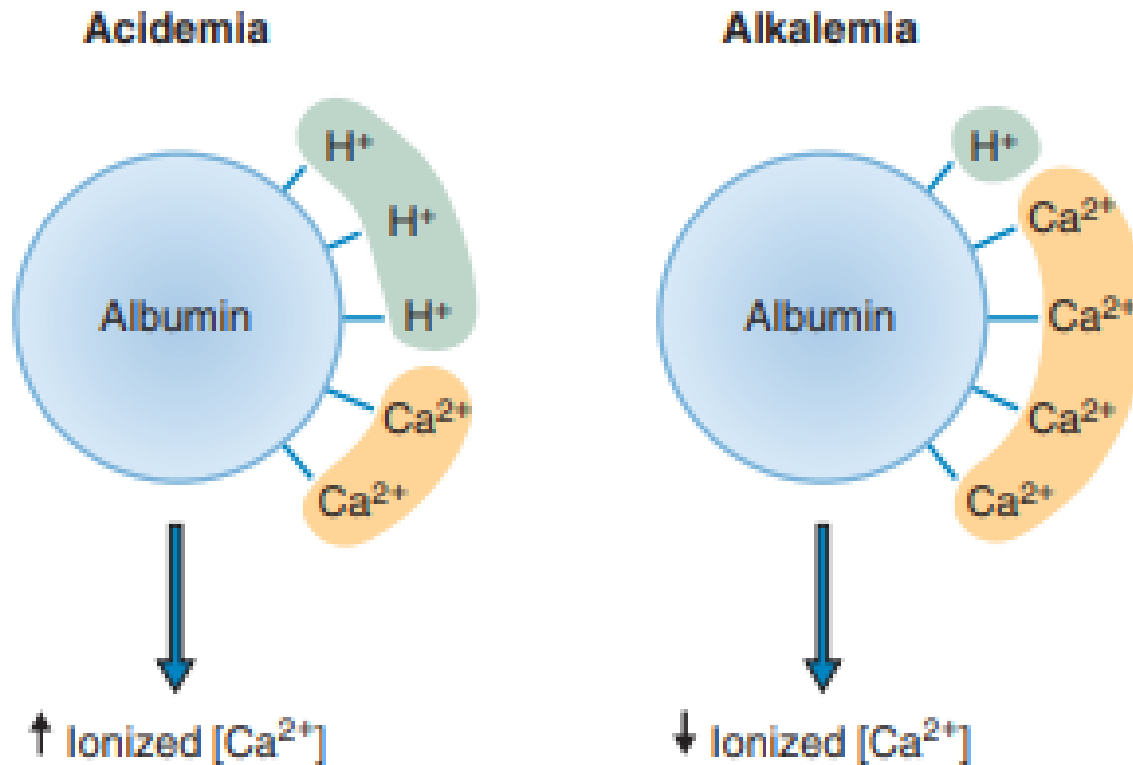
Slowly exchangeable pool (Stable pool)

- Bone is composed of organic matrix mostly collagen fibers 90 to 95%
- The rest is ground substance – ECF, proteoglycans, chondroitin sulfate & hyaluronic acid

Bone salts are deposited in the organic matrix mostly Ca²⁺ & PO₄, MgSO₄

Factors that alter – blood Ca^{2+} levels

At physiologic pH proteins are negatively charged



Changes in anions in plasma is also going to alter the plasma Ca^{2+} levels

Note: Effect of hyperventilation?

Factors that alter – blood Ca^{2+} levels

- Increased or decreased plasma protein alters total Ca^{2+} levels
- For a change in 1g/ dl of albumin concentration there is change 0.8mg/ dl of Ca^{2+} concentration

- Nephrotic syndrome
- Malnutrition
- Liver disease



↓ total Ca^{2+} levels

- Multiple myeloma



↑ total Ca^{2+} levels

In clinical lab evaluations it is important to measure proteins simultaneously with Ca^{2+} & PO_4 level

Phosphate distribution & balance

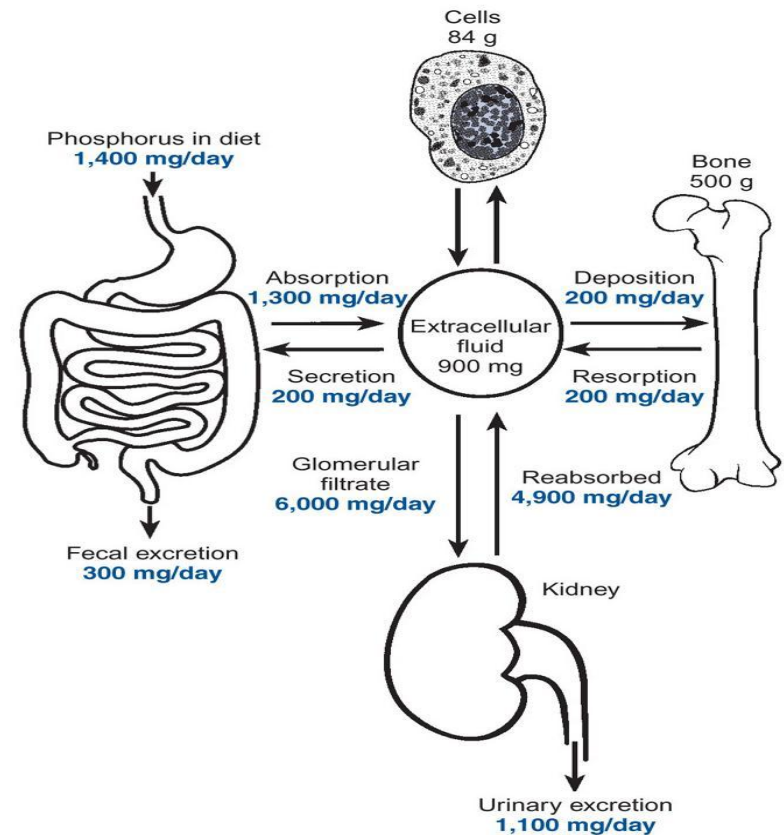
- Most of the dietary phosphate is absorbed. Serum concentration is 2.4 – 4.1 mg/dl

Phosphate distribution

86% is bone
- in elemental
form

- 13 – 14% in ICF
- organic form
- **very little in ECF**
- inorganic form

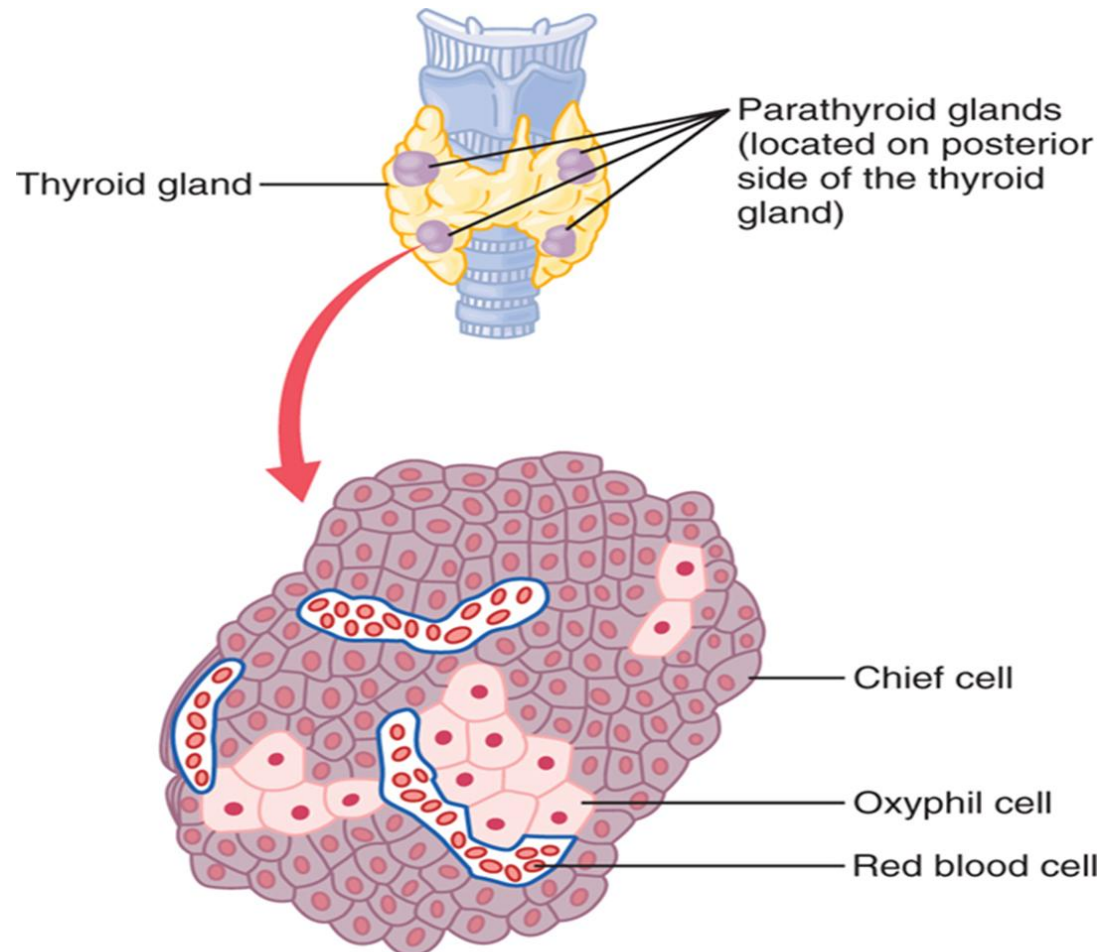
Typical daily exchanges of phosphorus



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Hormonal control

- Ca^{2+} homeostasis is mainly regulated by parathyroid (PTH) hormone, & calcitriol or vitamin D_3 , calcitonin
- These hormones regulate via 3 organs, the intestines, kidneys & the bone
- Calcitonin is not important for Ca^{2+} regulation in humans, but important in fetus

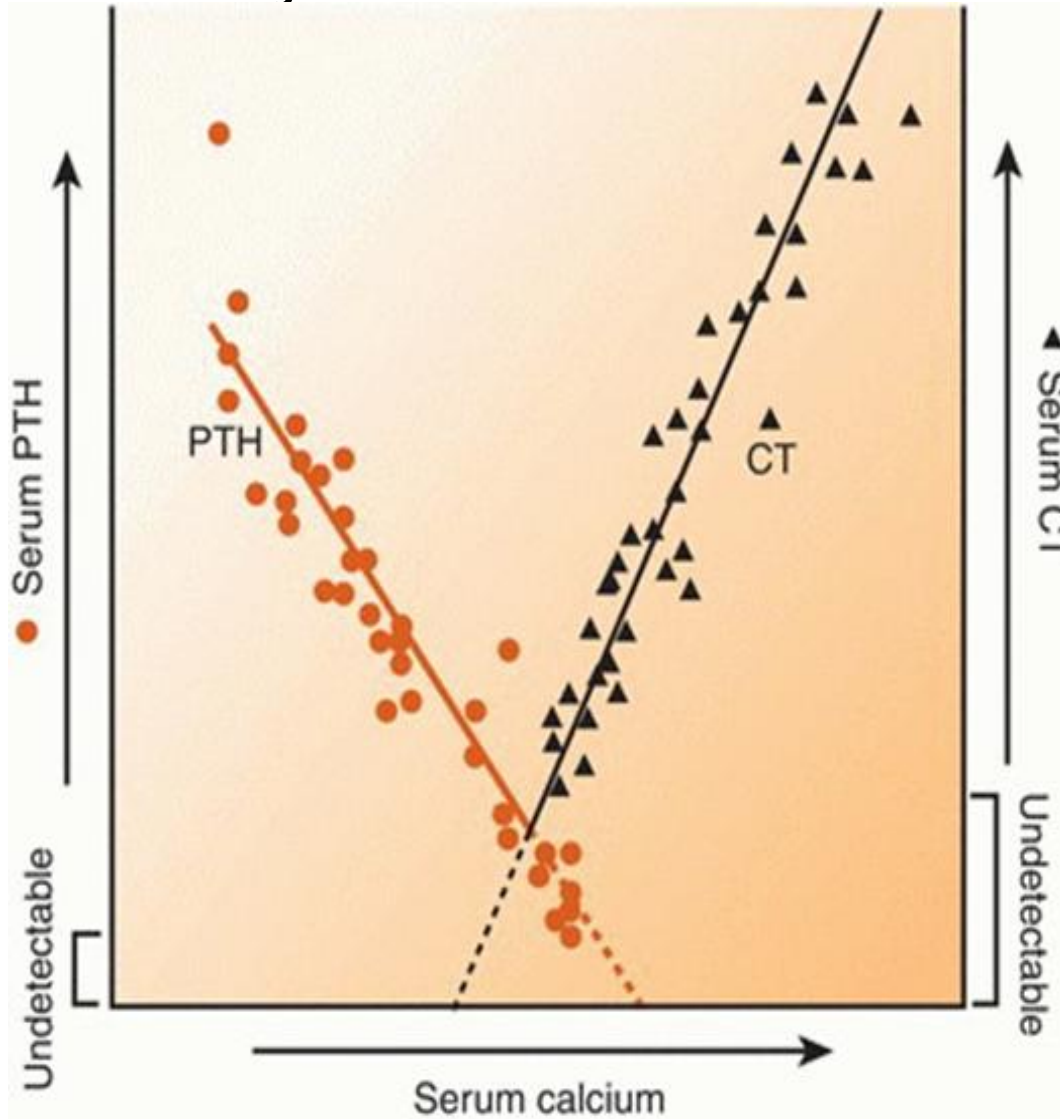


Hall: Guyton and Hall Textbook of Medical Physiology, 12th Edition
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Calcitonin

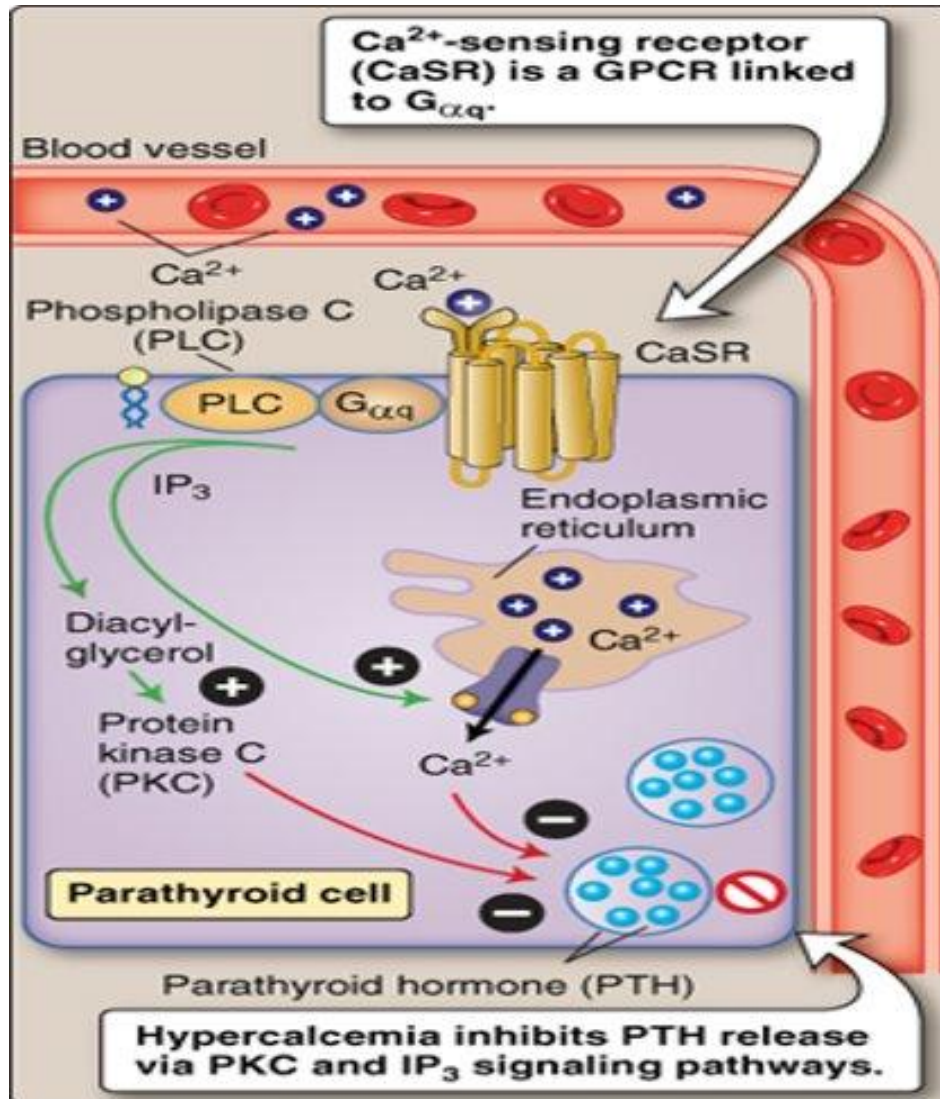
- Is secreted by C cells of the thyroid gland. It is a 32 AA peptide, has a half-life of 5 mins and secreted in response to hypercalcemia.
- Has opposite effect of PTH to reduce plasma Ca^{2+} by decreasing bone resorption by osteoclasts and increasing urinary Ca^{2+} excretion. **Helps in bone formation specially in fetus & neonates**
- In humans, calcitonin has weak effects of Ca^{2+} homeostasis, and neither the absence nor the excess of calcitonin causes a clinical defect in Ca^{2+} or phosphate homeostasis. However, it does have clinical applications:
 - Calcitonin is a tumor marker for medullary thyroid cancer, which derives from thyroid C cells.
 - Calcitonin can be used as an adjunctive therapy in hypercalcemia and in Paget's disease of the bone. Additionally, the use of calcitonin as a therapy is a classic example of tachyphylaxis, the development of rapid tolerance to drugs following repeated administration.

Parathyroid Hormone



- The primary physiologic stimulus for PTH secretion is a decrease in plasma calcium.
- The net effect of PTH is to increase the flow of calcium into plasma and return the plasma calcium concentration toward normal.
- Calcitonin (CT) secretion is stimulated by an increase in plasma calcium
- The net effect of CT is to promote calcium deposition in bone

Mechanism of PTH suppression



- When the extracellular Ca^{2+} concentration is increased, Ca^{2+} binds to the Ca^{2+} sensing receptor and activates phospholipase C. Activation of phospholipase C leads to increased levels of $\text{IP}_3/\text{Ca}^{2+}$, which inhibits PTH secretion.
- When extracellular Ca^{2+} is decreased, there is decreased Ca^{2+} binding to the receptor, which stimulates PTH secretion.

Parathyroid Hormone

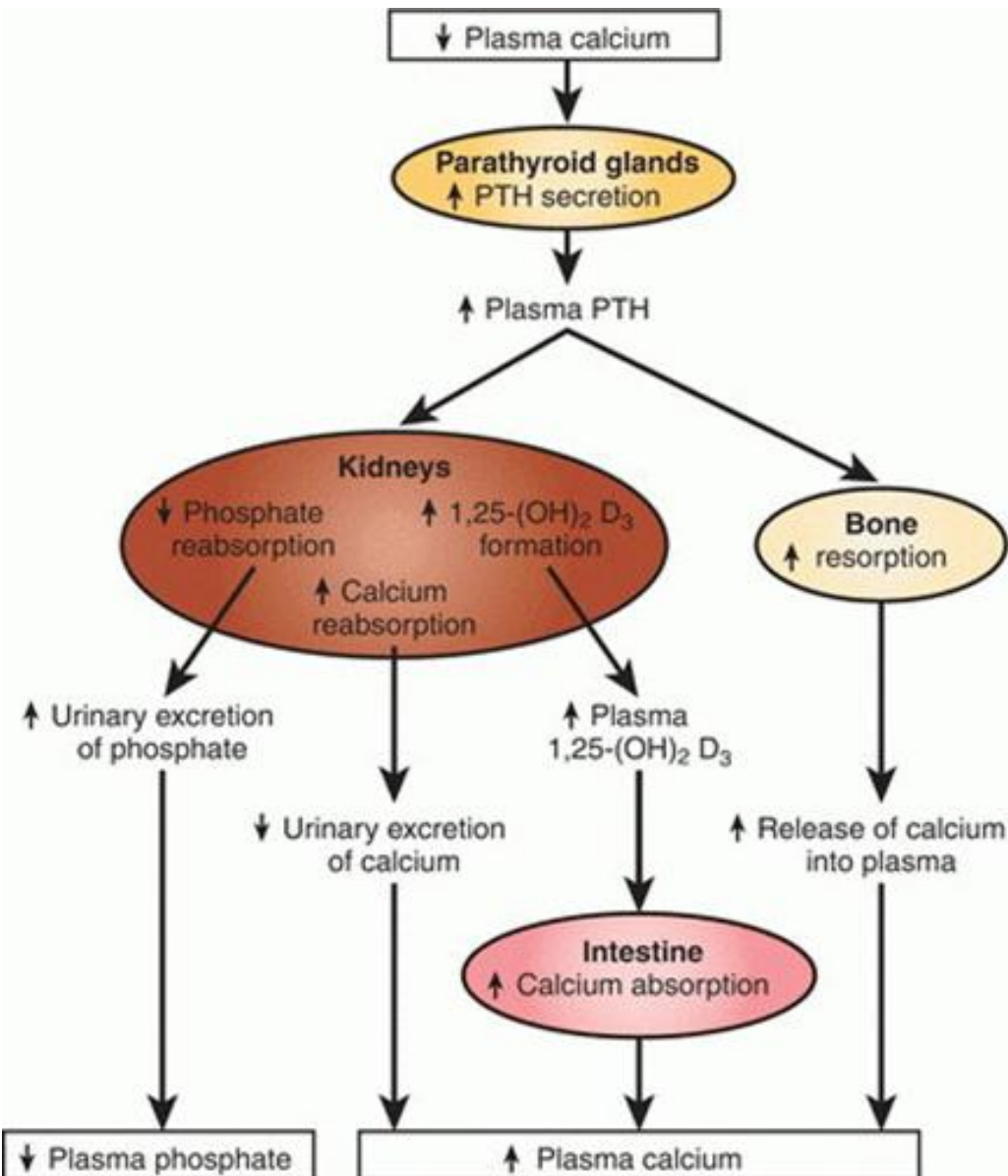
- PTH is an 84 amino acid peptide
- Is synthesized & stored in secretory granules
- It has a short half life 2- 4 minutes
- Rapidly removed from the kidney
- PTH is the most important regulator of ECF Ca^{2+}

Mechanism of PTH suppression

- Free Ca^{2+} levels are sensed by a calcium-sensing receptor (CaSR) expressed by parathyroid chief cells, which synthesize PTH.
- CaSR is a GPCR encoded by the *CASR* gene.
- When CaSR binds Ca^{2+} , it initiates Ca^{2+} release from intracellular stores and activates protein kinase C via $\text{G}_{\alpha q}$ and the IP_3 signaling system

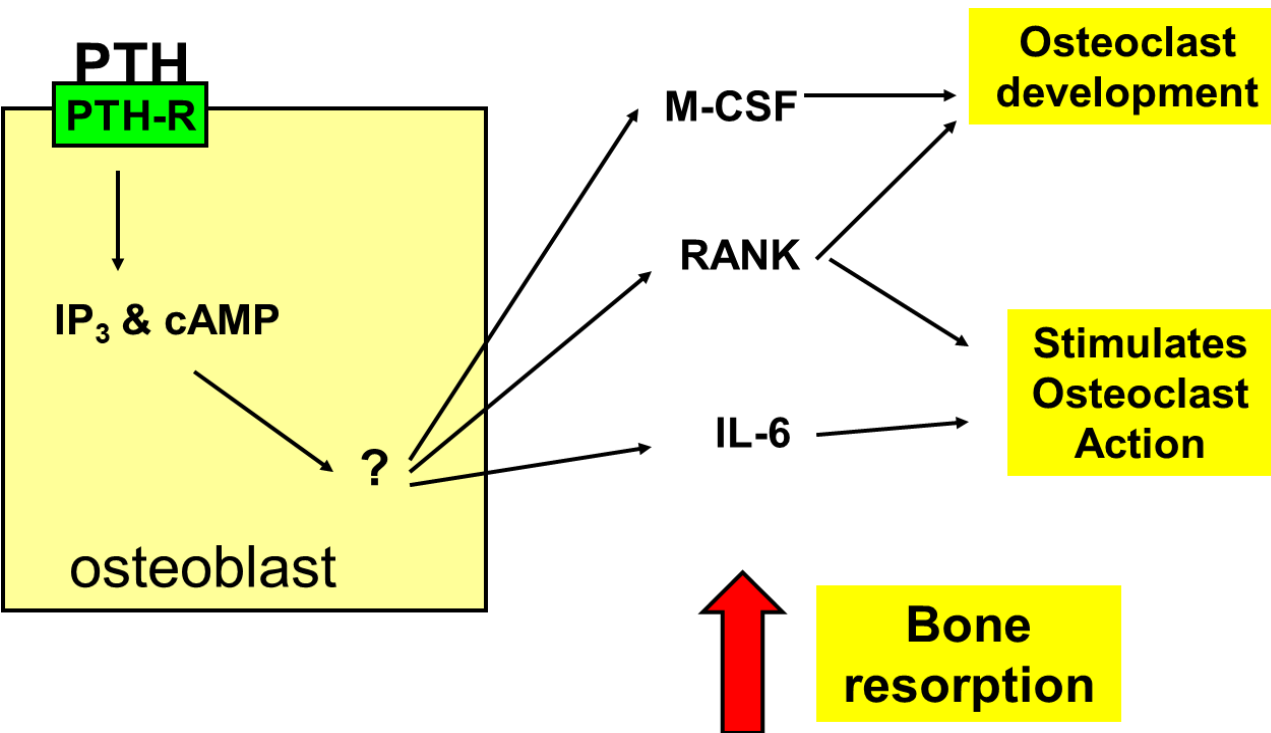
Action of PTH on Kidney, Bone, and Intestine

Effects on calcium and phosphate metabolism



- ↑ renal tubular reabsorption of Ca^{2+} in the distal tubules
- ↓ renal tubular reabsorption of PO_4
- Indirectly increases intestinal absorption of Ca^{2+}
 - via vitamin D

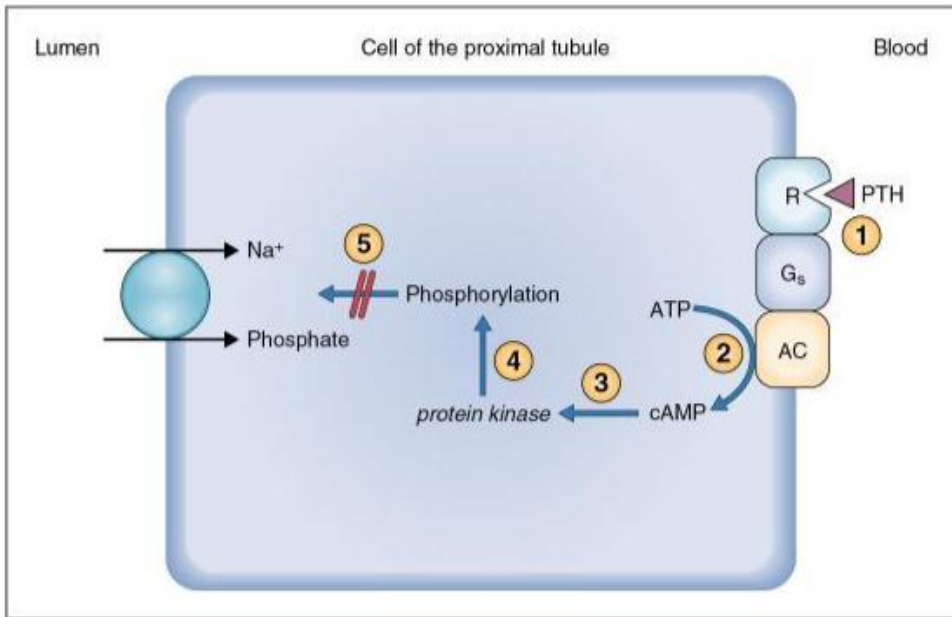
Action of PTH on bone



- PTH activates osteoclasts to increase bone resorption and the delivery of calcium from bone into plasma.
- PTH stimulates the maturation of immature osteoclasts into mature, active osteoclasts.

Note: Monoclonal antibodies against RANK-L are used to treat osteoporosis; the rate of organic bone matrix resorption can be assessed by measuring urinary excretion of **hydroxyproline**

Action of PTH on kidney

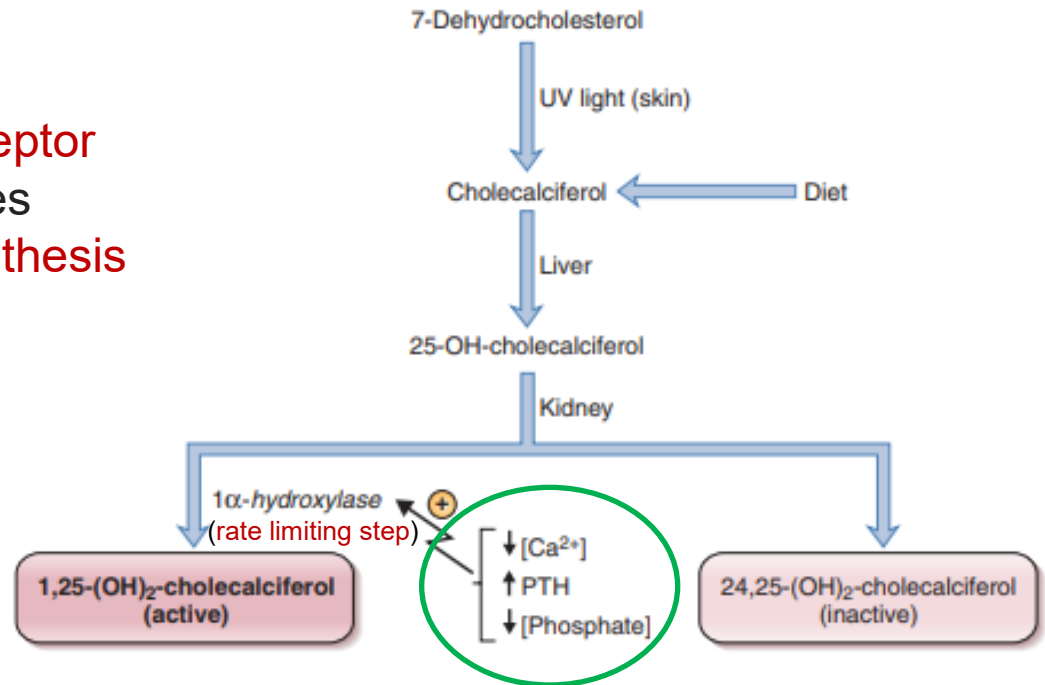


Mechanism of action of PTH on the renal proximal tubule

- $\uparrow$ renal tubular reabsorption of Ca^{2+} in the distal tubules
- $\downarrow$ renal tubular reabsorption of PO_4 in the proximal tubules
- Indirectly increases intestinal absorption of Ca^{2+}
 - via vitamin D (PTH regulates the enzyme 1α -hydroxylase that converts 25-hydroxycholecalciferol to 1,25-dihydroxycholecalciferol)

Vitamin D and PTH

- Vitamin D is a group of steroidal substances
- Binds to a **nuclear vitamin D receptor (VDR)**. VDR complexes increases **gene expression and protein synthesis**
- Vitamin D₃ (cholecalciferol) is important for Ca²⁺ homeostasis
 - formed in the skin
 - can be obtained from diet

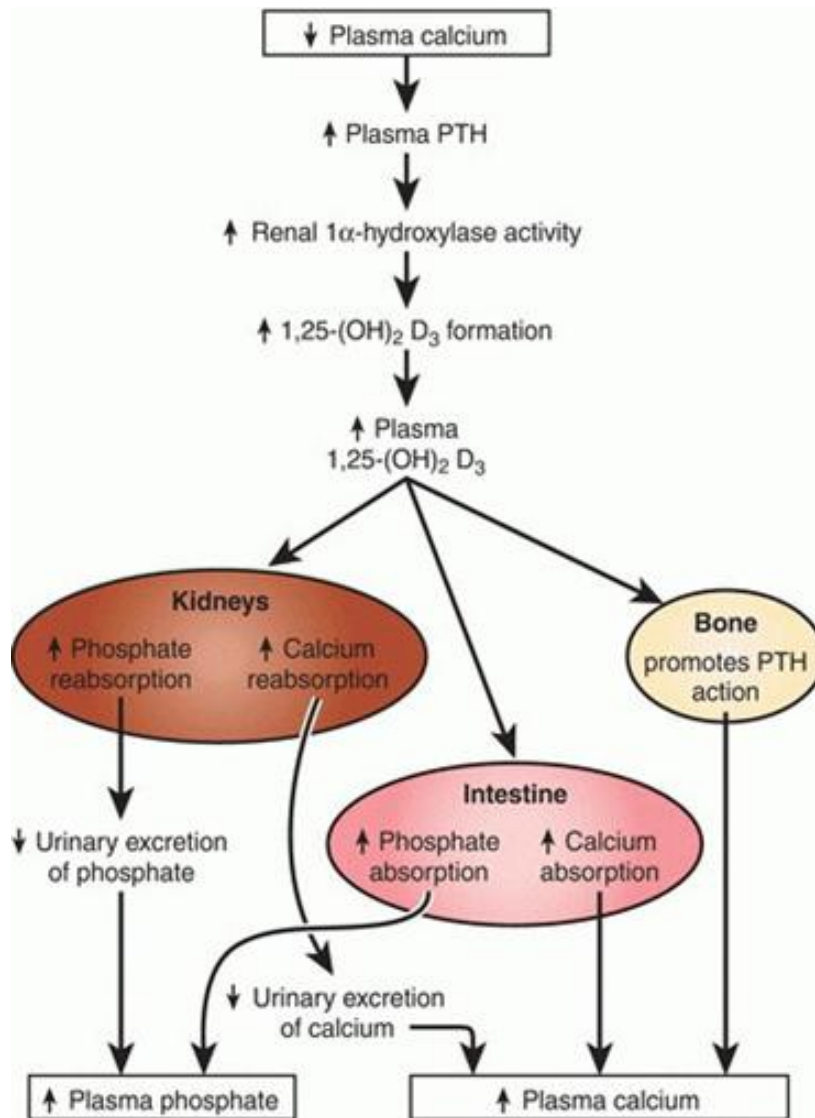


Cutaneous synthesis of calcitriol

Note: Patients with chronic renal failure are often deficient in vitamin D because the diseased kidneys are unable to sufficiently convert inactive vitamin D to its active form

Vit D and the action on Kidney, Bone, and Intestine

Effects of $[1,25-(\text{OH})_2 \text{D}_3]$ on calcium and phosphate metabolism

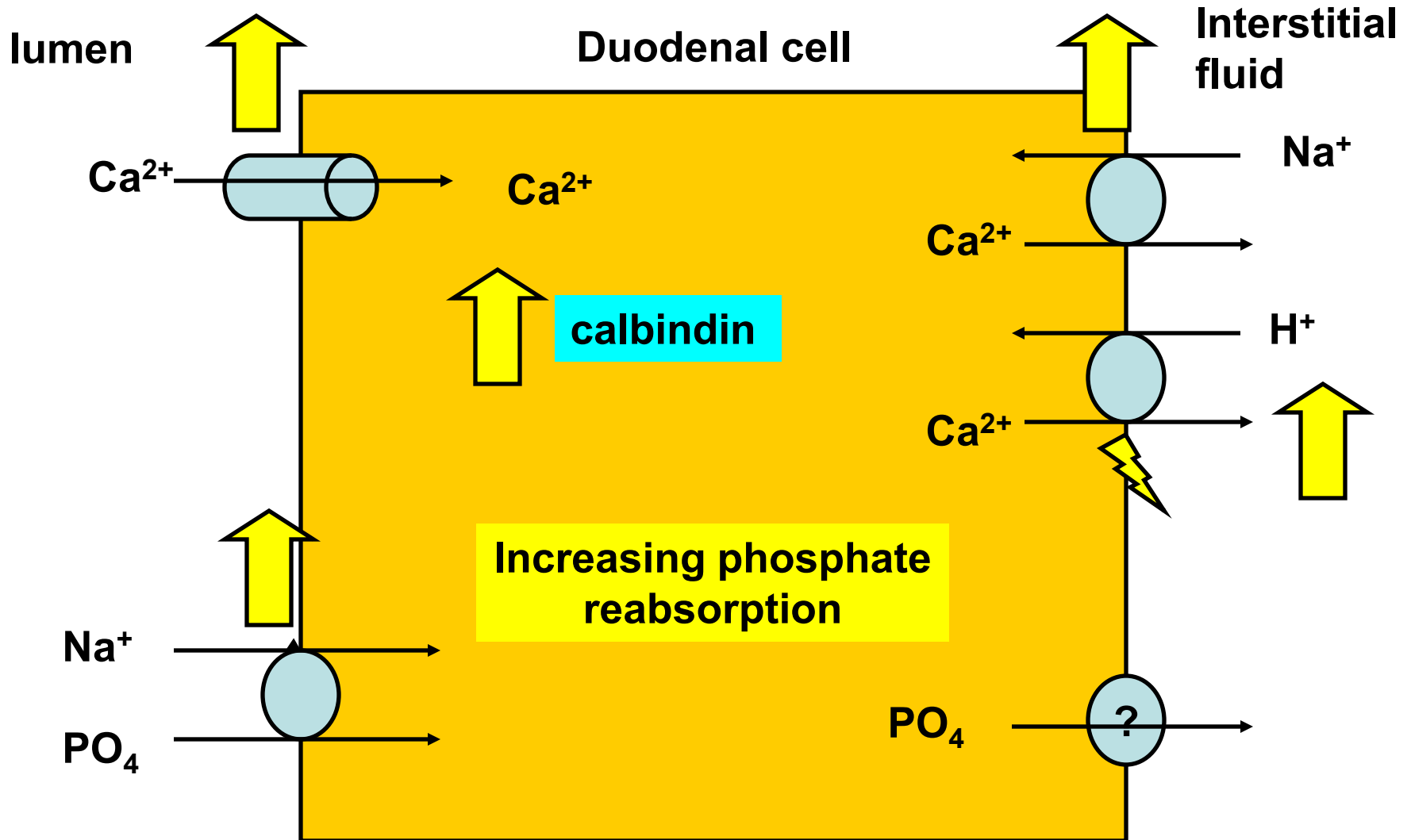


- In the kidneys, effect of $1,25-(\text{OH})_2 \text{D}_3$ is weak.
- In bones, effect of $1,25-(\text{OH})_2 \text{D}_3$ is indirect (by promoting PTH action).
- In the intestine, effect of $1,25-(\text{OH})_2 \text{D}_3$ is **direct** → stimulates calcium and phosphate absorption by the small intestine → **increasing plasma concentrations of both ions**.
- This effect is mediated by **increased production of calcium transport proteins (calbindin)** resulting from gene transcription events and usually requires several hours to appear.

Vit D and the action on GIT

Maintaining Low intracellular Calcium levels

Increasing Calcium Reabsorption By GIT



Summary of the functions of Parathyroid hormone (PTH)

- PTH is released in response to **hypocalcemia** and **increases plasma calcium** and **lowers plasma phosphorus** by the following mechanisms:
- In the bones, PTH activates osteoclasts to increase bone resorption, releasing calcium and phosphorus.
- In the kidneys, PTH **decreases proximal tubular reabsorption of phosphate** and **increases calcium reabsorption in the distal convoluted tubule and collecting duct**.
- PTH **activates 1-alpha-hydroxylase** in the kidney, which converts 25-hydroxyvitamin D to 1,25-dihydroxyvitamin D (Calcitriol, active form of Vit D). Calcitriol increases calcium and phosphorus absorption in the small intestine. This is an indirect action of PTH.

Disorders of Calcium & Vitamin D imbalance

Features

Hypocalcemia

- **Increases membrane excitability**
 - Paresthesia
 - Hypocalcaemic tetany
 - Chvostek's sign twitching of facial muscles on tapping facial nerve*
 - Trousseau's sign when hand goes into tetany on inflating the BP cuff to systolic level*
 - **EKG shows prolonged QT interval**



Induces ↑PTH

Hypercalcemia

- Nephrogenic DI – polyuria, polydipsia
- **PUD** (peptic ulcer disease) due to direct stimulation of parietal cells
- Urolithiasis or kidney stones (hypercalciuria-may lead to kidney stone)
- Neurologic symptoms (hyporeflexia, lethargy, coma, and death).
- **EKG shows shortened QT interval**



↓PTH secretion

* <https://www.youtube.com/watch?v=kvmwsTU0InQ>

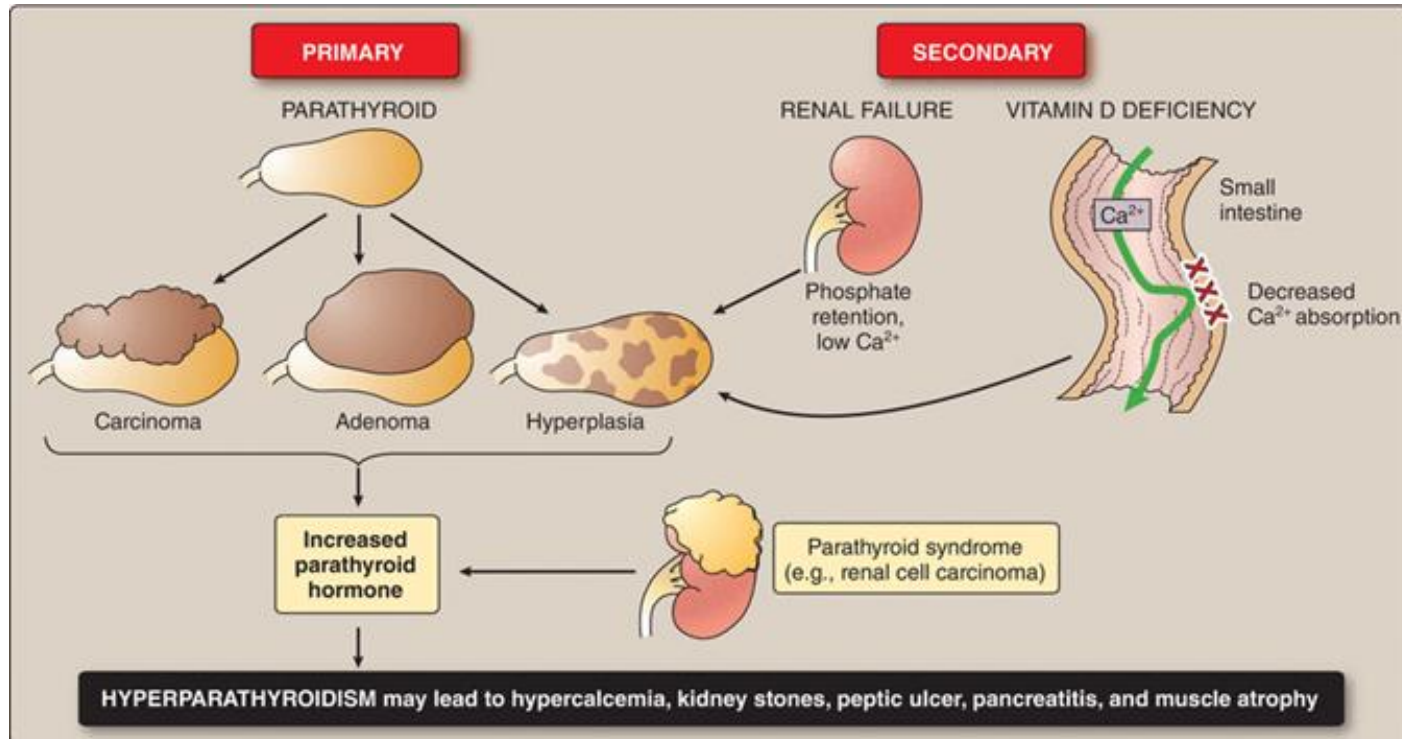
Chvostek's Sign – contraction of ipsilateral facial muscles with the facial nerve is tapped just anterior to the ear. Contraction of the mouth alone occurs in 10-30% of normal individuals.



Trousseau's Sign – induction of carpopedal spasm by inflation of bp cuff above SBP for 3 min. Characterized by adduction of thumb, flexion of MCP joints, extension of interphalangeal joints, flexion of the wrist. More specific than Chvostek's.



Primary and Secondary hyperparathyroidism



Major pathogenic pathways leading to primary and secondary hyperparathyroidism

Lippincott Illustrated Reviews: Integrated Systems, 1e, 2016

Primary hyperparathyroidism

- It is a clinical condition in which parathyroid gland secretes excessive PTH which results in hypercalcemia and other clinical features of excess PTH.
- Most common causes are
 - parathyroid adenoma,
 - parathyroid carcinoma,
 - parathyroid hyperplasia.
- **Excess PTH causes**
 - **↑ serum Ca^{2+}**
 - **↓ serum phosphate (due to phosphaturia)**
- Clinical features:
 - polyuria, calciuria and kidney stones
 - prone for peptic ulcer disease
 - develop cystic lesions in the bone
- EKG shows shortened QT interval
- **Patients are said to have “stones, bones, groans, and moans”**

Secondary Hyperparathyroidism

- It is a clinical condition in which the parathyroid gland secretes excessive PTH in response to a low serum calcium.
- Reduced negative feedback on the gland because of a low serum calcium stimulates the gland to secrete excess PTH. Increased serum PTH is **secondary** to a low serum calcium.
- Major causes of a low serum calcium that initiates this condition are:
 - Vitamin D deficiency (Either dietary deficiency or poor absorption in the GIT)
 - Chronic Kidney disease
- **Vitamin D deficiency** (Rickets in children and Osteomalacia in adults) results in:
 - ↓ serum Ca^{2+} and ↓ urine Ca^{2+}
 - ↑ serum PTH
 - ↓ Serum phosphate but normal to ↑ urine phosphate levels
- **Chronic kidney disease** results in:
 - ↑ Serum Phosphate (hyperphosphatemia)
 - ↓ serum Ca^{2+} and ↓ urine Ca^{2+}
 - ↑ Serum PTH

(In Chronic kidney disease (CKD), the serum Phosphate levels are elevated because of a reduced GFR and decreased urinary excretion of phosphate).

Primary hypoparathyroidism

- Causes
 - **accidental** removal of glands during **thyroid** or head & neck **surgery**
- Characterized by
 - **low Ca^{2+} & high plasma phosphate levels**
 - **tetany** (uncontrollable muscle contractions) is due to $\uparrow$ muscle excitability
- EKG shows prolonged QT interval
- $\downarrow \text{Ca}^{2+} < 6\text{mg/dl}$ can lead to death

Pseudo-hypoparathyroidism

Caused by a defective G-protein linked to the PTH receptor in the target organs (kidney & bone) which makes tissue resistant to PTH action.

- a. Patients present with low plasma $[Ca^{2+}]$ and high [phosphate] because the normal tissue responses to PTH do not occur.
- b. PTH levels are high due to hypocalcemia but tissue resistance to PTH prevents a physiological response.

Rickets & Osteomalacia

- Causes due to Vitamin D deficiency
- There are ↓ levels of Ca^{2+} & PO_4 in ECF
- Characterized by demineralization of the osteoid matrix
 - in children it causes rickets
 - in adults it leads to osteomalacia

Rachitic Rosary – Vitamin D deficiency



Soft tissue overgrowth at costochondral junctions.

Bonus Question

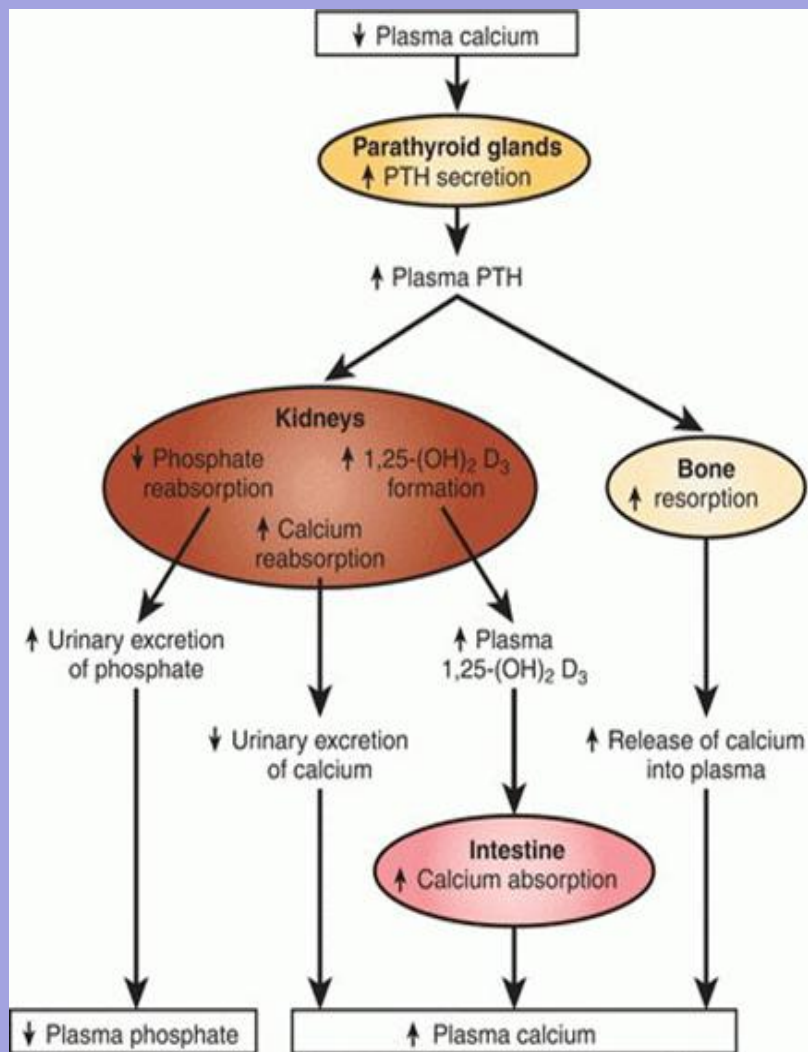
Vitamin D is a product of cholesterol metabolism. Which of the following is not either directly or indirectly involved in formation of active vitamin D?

- A. Bone
- B. Skin
- C. Kidney
- D. Liver
- E. Both kidney and liver

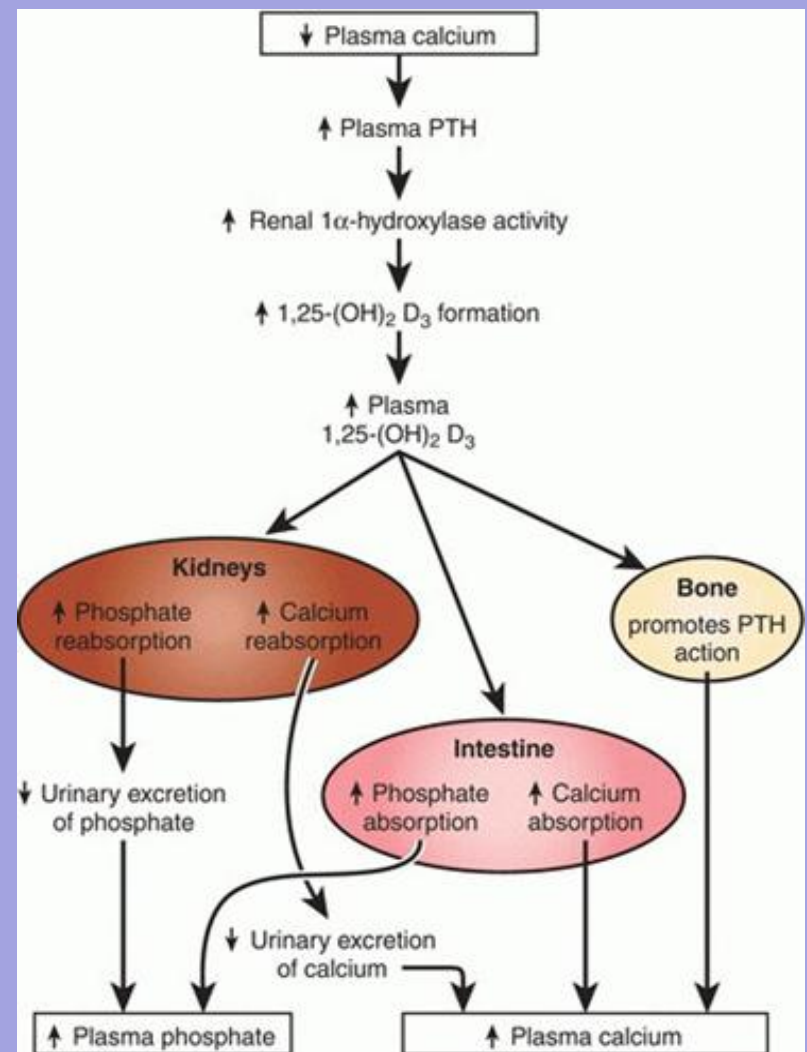
The correct answer is A. Skin, kidney, and liver can all be involved in forming the active metabolite of vitamin D, 1,25-dihydroxycholecalciferol. Bone does not form this hormone but is a target for its actions.

Key Concepts

- PTH release is controlled by circulating Ca^{2+} levels and is under negative feedback regulation by Ca^{2+}
- In the kidney, PTH increases renal Ca^{2+} reabsorption, increases the activity of 1α -hydroxylase (which mediates the final activation step in the synthesis of vitamin D), and decreases Pi reabsorption
- In bone, PTH increases osteoclast-mediated bone resorption indirectly through stimulation of osteoblast activity
- Calcitonin decreases bone resorption and lowers plasma Ca^{2+} levels
- Synthesis of active vitamin D involves hydroxylation in the liver (C-25) and kidney (C-1)
- Vitamin D increases intestinal Ca^{2+} absorption, facilitates renal Ca^{2+} reabsorption, and increases bone turnover



Effects of parathyroid hormone (PTH) on calcium and phosphate metabolism



Effects of 1,25-dihydroxycholecalciferol [1,25-(OH)₂ D₃] on calcium and phosphate metabolism

References

- Rhoades RA, Bell DR. Medical Physiology: Principles for Clinical Medicine, 6e, Chapter 35 [Endocrine Regulations of Calcium, Phosphate, and Bone Homeostasis | Medical Physiology: Principles for Clinical Medicine, 6e | Medical Education | Health Library \(oclc.org\)](#)
- Costanzo, L. S. (2021). *Costanzo Physiology* (7th ed.). Elsevier Limited (UK)., 449-460 [ClinicalKey Student: Costanzo Physiology \(elsevier.com\)](#)

Thank you

For question and clarifications
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